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EX: 3 LOG TRANSFORMATION OF INTERNATIONAL AIRLINE PASSENGER DATA

AIM

To apply a logarithmic transformation to the International Airline Passenger time series and compare the original and transformed data.

ALGORITHM

1. Start the program.
2. Import Pandas, NumPy, Matplotlib, and seasonal decomposition functions.
3. Load the AirPassengers.csv dataset.
4. Convert the Month column to datetime and set it as the index.
5. Display and plot the original passenger series.
6. Apply logarithmic transformation using np.log().
7. Display and plot the log-transformed series.
8. Calculate and compare the variance before and after transformation.
9. Plot the original and log-transformed series for comparison.
10. Perform seasonal decomposition on the log-transformed series with period 12.
11. Observe the change in variance and seasonal behavior.
12. Stop the program.

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PROGRAM

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset
df = pd.read_csv("AirPassengers.csv")

# Convert Month to datetime
df["Month"] = pd.to_datetime(df["Month"])

# Set Month as index
df.set_index("Month", inplace=True)

# Display original data
print("Original Data:")
print(df.head())

# Plot original series
plt.figure(figsize=(12, 5))
plt.plot(df.index, df["Passengers"], marker="o")

plt.title("International Airline Passengers")
plt.xlabel("Year")
plt.ylabel("Passengers (Thousands)")
plt.grid(True)
plt.show()

# Apply log transformation
df["Log_Passengers"] = np.log(df["Passengers"])

print("\nLog Transformed Data:")
print(df.head())

# Plot log-transformed series
plt.figure(figsize=(12, 5))
plt.plot(df.index, df["Log_Passengers"], marker="o")

plt.title("Log-Transformed Airline Passenger Data")
plt.xlabel("Year")
plt.ylabel("Log(Passengers)")
plt.grid(True)
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```
plt.show()

# Compare original and transformed variance
print("\nVariance Comparison:")
print("Original Variance:", df["Passengers"].var())
print("Log Transformed Variance:", df["Log_Passengers"].var())

# Compare original and log-transformed series
fig, axes = plt.subplots(2, 1, figsize=(12, 8))

axes[0].plot(df.index, df["Passengers"])
axes[0].set_title("Original Airline Passenger Series")
axes[0].set_ylabel("Passengers")
axes[0].grid(True)

axes[1].plot(df.index, df["Log_Passengers"])
axes[1].set_title("Log-Transformed Airline Passenger Series")
axes[1].set_ylabel("Log(Passengers)")
axes[1].set_xlabel("Year")
axes[1].grid(True)

plt.tight_layout()
plt.show()

# Seasonal decomposition of log-transformed data
result = seasonal_decompose(
    df["Log_Passengers"],
    model="additive",
    period=12
)

result.plot()
plt.show()
```

OUTPUT

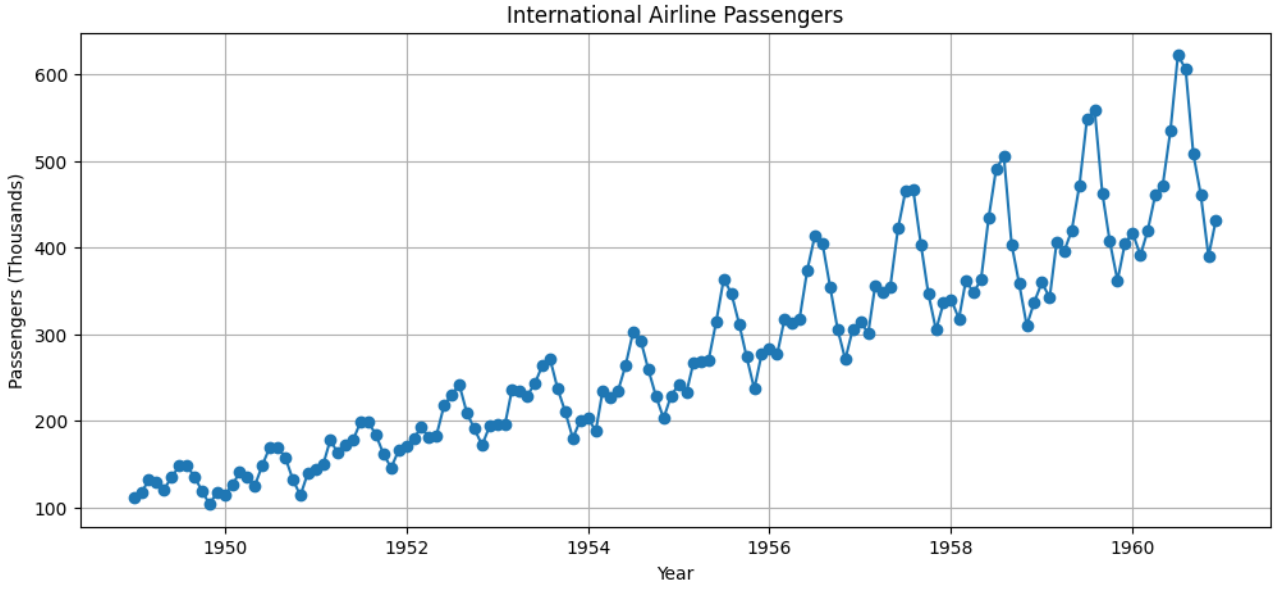
```
Original Data:
            Passengers
Month
1949-01-01         112
1949-02-01         118
1949-03-01         132
1949-04-01         129
1949-05-01         121

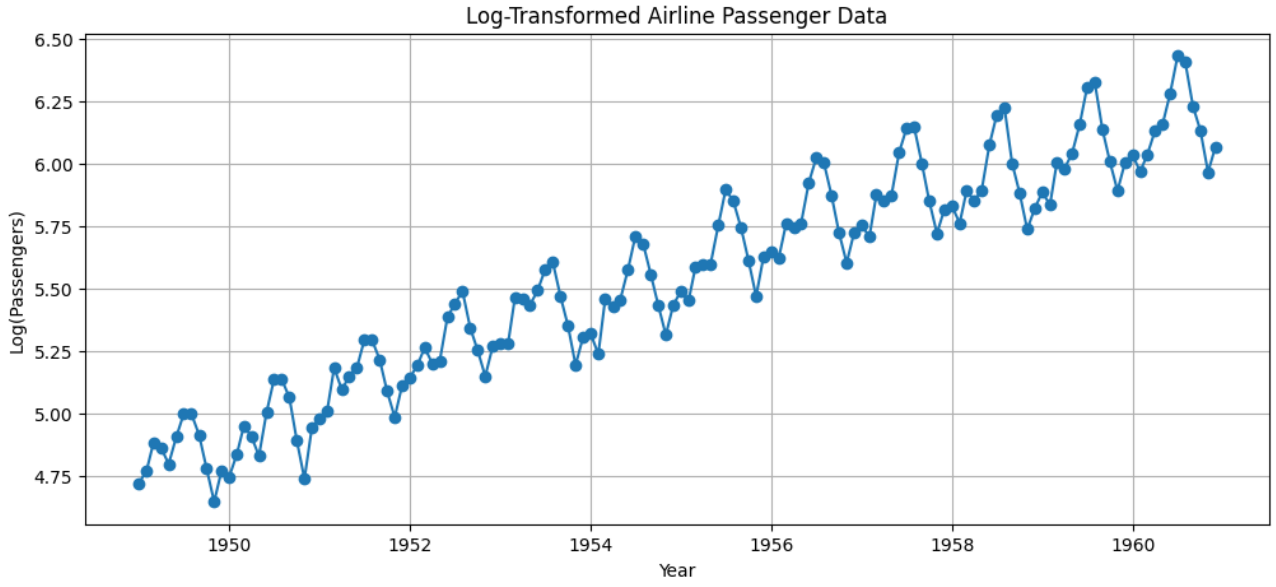
Log Transformed Data:
            Passengers  Log_Passengers
Month
1949-01-01         112        4.718499
1949-02-01         118        4.770685
1949-03-01         132        4.882802
1949-04-01         129        4.859812
1949-05-01         121        4.795791

Variance Comparison:
Original Variance: 14391.917200854701
Log Transformed Variance: 0.19488377448470776
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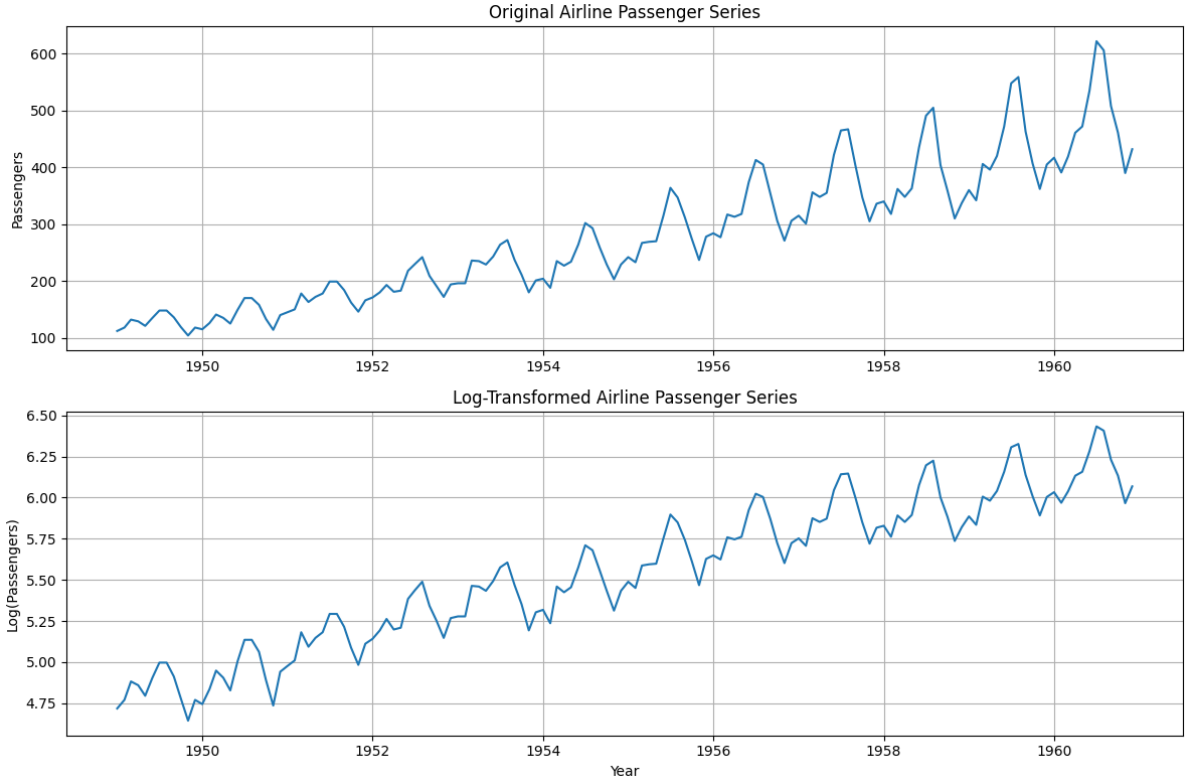
GRAPHS


International Airline Passengers
Passengers (Thousands)
100
200
300
400
500
600
1950
1952
1954
1956
1958
1960
Year

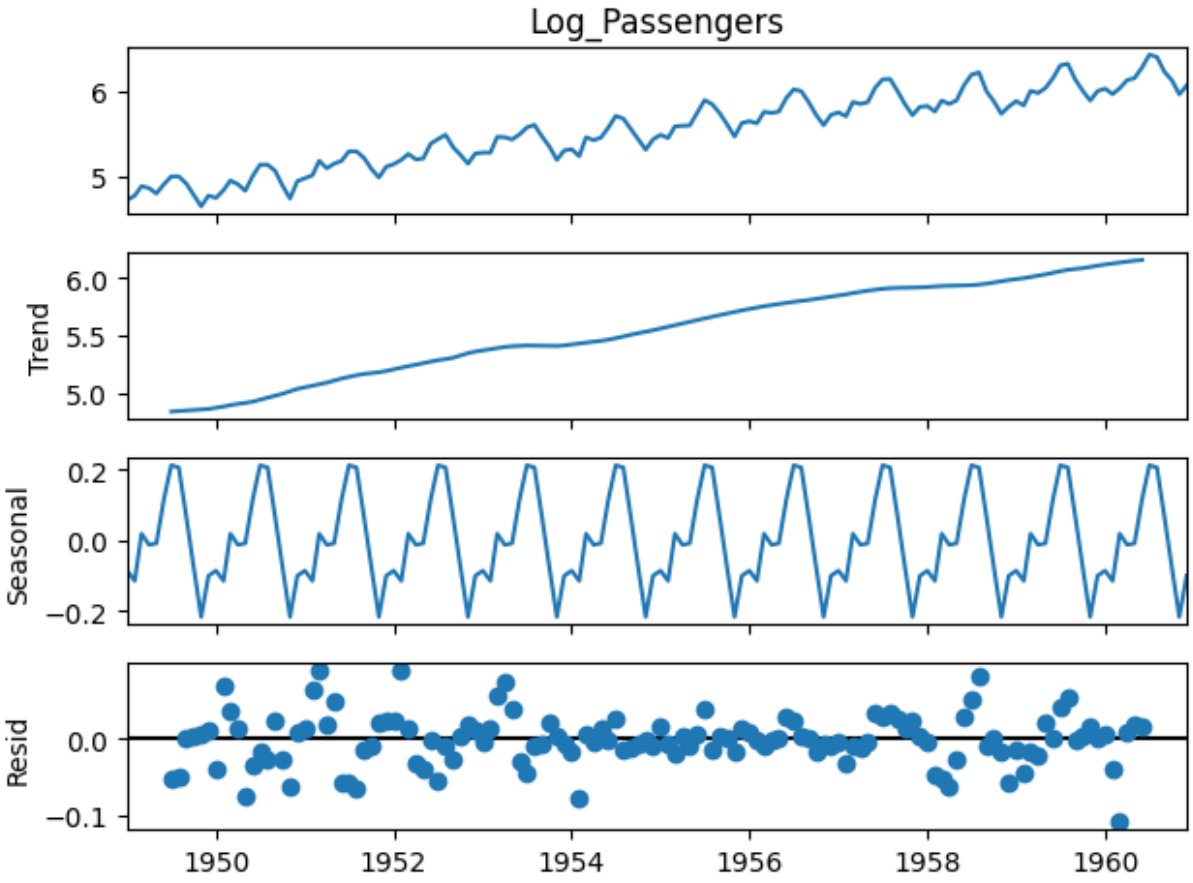

Log-Transformed Airline Passenger Data
Log(Passengers)
4.75
5.00
5.25
5.50
5.75
6.00
6.25
6.50
1950
1952
1954
1956
1958
1960
Year

DATE:

GRAPHS



SEASONAL DECOMPOSITION



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OBSERVATION

The original passenger series shows increasing variation as the passenger count rises. After applying the logarithmic transformation, the scale is compressed and the variance is greatly reduced. The transformed series has more stable variability, while the seasonal pattern remains clearly visible.

RESULT

Thus, the logarithmic transformation was successfully applied to the International Airline Passenger time series. The original and transformed series were compared, variance was reduced, and seasonal decomposition was performed successfully.